

RAPHAEL LEITE DA COSTA NASCIMENTO

Junior Machine Learning Engineer
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Languages Portuguese - Native — English - Intermediate

EXPERIENCE

Active Member - TAIL (UFPB) 08/2022 - Present
Developed **Artificial Intelligence** projects in a multidisciplinary team environment. Implemented an **LLM-based agent with RAG** (Retrieval-Augmented Generation), including knowledge base structuring, document indexing, and context retrieval for more accurate response generation. Participated in a **Computer Vision** project for automatic fire extinguisher detection in floor plans, involving image preprocessing and model validation.

Software Engineer Intern - Synchro [São Paulo - SP] 10/2022 - 11/2024
Developed and maintained internal applications using **Spring Boot** and **React.js**, focusing on system integration and optimization of corporate tax processes. Designed and implemented **REST APIs**, integrated with relational databases, and built reusable front-end components. Worked in an Agile environment, collaborating with cross-functional teams (product, design, and QA) on requirements gathering, business rule validation, and application performance improvements.

Undergraduate Researcher - Networking Laboratory (LAR/UFPB) 09/2021 - 10/2022
Conducted security analysis and testing on BLE devices using the Sneytnooth framework and nRF52840 Dongle hardware in a Linux environment. Performed crash, deadlock, and security bypass attacks on IoT devices, analyzing BLE protocol vulnerabilities. Developed Python automation scripts to manipulate packets and assess security risks. Contributed to identifying flaws that could compromise the privacy and integrity of connected devices.

EDUCATION

Bachelor's Degree in Data Science and Artificial Intelligence
Universidade Federal da Paraíba (UFPB) - GPA: 8.89/10.0 05/2019 - 11/2025

TOOLS

Languages	Java, Python, JavaScript, C/C++, SQL
Frameworks	PyTorch, TensorFlow, Keras, LangChain, LangGraph
Tools	Git, Linux, Docker, Gradle

EXTRACURRICULAR ACTIVITIES

Teaching Assistant - Introduction to Programming - UFPB 03/2021 - 07/2021
Assisted first-semester students in programming logic, problem-solving, and provided support for coursework and projects.

PUBLICATION

NASCIMENTO, Raphael L. C.; LEAL, Ryan M. S.; FONSECA, Iguatemi E.
Experimental Demonstration of Vulnerabilities in Internet of Things Devices.
XXII Brazilian Symposium on Information Security and Computer Systems (SBSeg), 2022.

CERTIFICATIONS

Google AI Essentials - Coursera 12/2024

HONORS AND AWARDS

Best Short Paper - Undergraduate Research and Graduation Workshop - SBSeg 2022 09/2022